Recurrance Relations

A relation where a_n (the nth term of the sequence $\{a_i\}_{i\geq 0}$) depends on the prior terms $a_0, a_1, a_2, \ldots, a_{n-1}$, is called a recurrance relation. Such a relation is also called a difference equation which is the discrete counterpart of differential equation. Our attention is to analyze linear recurrance relation with constant coefficient.

Order and degree of a recurrance relation:-

Let $c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} = f(n)$, with $n \geq k$ be a recurrance relation (with constant coefficient) of order k, where $c_0 (\neq 0), c_1, c_2, \ldots, c_k (\neq 0)$ be real number constants, and $c_i \in \mathbb{R}$, $i = 0, 1, 2, \ldots, k$.

This is a linear recurrance relation. The highest power of the subscripted variables $(a_n, a_{n-1}, \ldots, a_{n-k})$ is called the degree of the recurrance relation. Sometimes the sum of the powers of the product form of subscripted variables determined the degree of a recurrance relation.

The difference between the highest and lowest subscripts determine the order of a recurrance relation. In the above relation the order is $[n-(n-k)] = k$

When $f(n) = 0$ for all $n \geq 0$, the relation is called homogeneous; otherwise it is called nonhomogeneous recurrance relation.

★ What is the meaning of solution of a recurrance relation?

Solution of a recurrance relation essentially means that expressing a_n (n-th term of the sequence $\{a_i\}_{i\geq 0}$) as a function of n.

★ How to solve a linear recurrance relation with constant coefficient.

First we would like to discuss the solution methodology of any linear recurrance relation of the following form:

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} = 0 \quad (*)$$

This is called a homogenous recurrance relation as discussed above. We seek a solution of the form $a_n = dr^n$, where $d \neq$ and $r \neq 0$, and both d and r can be complex numbers.

Substituting $a_n = dr^n$, in the above recurrance relation $(*)$, we obtain

$$c_0 dr^n + c_1 dr^{n-1} + c_2 dr^{n-2} + \cdots + c_k dr^{n-k} = 0,$$

and using the assumption $d \neq 0$, $r \neq 0$, this equation gives

$$c_0 r^n + c_1 r^{n-1} + c_2 r^{n-2} + \cdots + c_k r^{n-k} = 0,$$

which is called the characteris equation (CE).

$$\Rightarrow c_0 r^k + c_1 r^{k-1} + c_2 r^{k-2} + \cdots + c_k = 0$$

The k roots of this CE determine the solution of the recurrance relation.

Depending on the nature of the roots of the CE. One can get different form of solution of the recurrance relation. These are discussed casewise below.

Case I: Distinct real roots

The CE $c_0 r^k + c_1 r^{k-1} + c_2 r^{k-2} + \cdots + c_k = 0$ have $r_1, r_2, \ldots, r_k$ distinct real roots. The recurrance relation has a general solution of the form:

$$a_n = d_1 r_1^n + d_2 r_2^n + \cdots + d_k r_k^n,$$

where the unknown constants $d_1, d_2, \ldots, d_k$ may be obtained using first k values of the recurrance relation $a_0, a_1, a_2, \ldots, a_{k-1}$.

[Example: Solve the second order homogeneous relation $\underbrace{F_{n+2} = F_{n+1} + F_n}_{\text{Fibonacci relation}}$, where $n \geq 0$, $F_0 = 0$, $F_1 = 1$.

Assume $F_n = d r^n$, $d \neq 0$, $r \neq 0$, $n \geq 0$.

Upon substitution $F_n = d r^n$ in the Fibonacci relation and using the assumption $d, r \neq 0$, we obtain the following CE: $r^2 - r - 1 = 0$, The characteristic roots are $r = (1 \pm \sqrt{5})/2$, so the general solution is

$$F_n = d_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + d_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$$

Solve for d_1, d_2 using $0 = F_0 = d_1 + d_2$ & $1 = F_1 = d_1\left(\frac{1+\sqrt{5}}{2}\right) + d_2\left(\frac{1-\sqrt{5}}{2}\right)$, which

gives $d_1 = \frac{1}{\sqrt{5}}$ and $d_2 = -\frac{1}{\sqrt{5}}$. Thus the general solution is given by

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right], \quad n \geq 0.$$

<u>Case II: Complex roots</u>

If the characteristic roots are complex numbers then then they must occur in pairs, and each pair of roots consists of a complex number and its conjugate complex number. The corresponding multiplicative constants are also consists of another complex number and its conjugate, so that the corresponding general solution becomes a real number.

Example: Solve the recurrance relation $a_n = 2(a_{n-1} - a_{n-2})$, where $n \geq 2$ and $a_0 = 1$, $a_1 = 2$.

Let $a_n = d r^n$, for $d, r \neq 0$, we obtain the CE $r^2 - 2r + 2 = 0$ whose roots are $r = 1 \pm i$. Consequently, the general solution has the form $a_n = d_1(1+i)^n + d_2(1-i)^n$, where d_1 and d_2 are arbitrary complex constants.

$$1+i = \sqrt{2}\left[\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right]$$

$$1-i = \sqrt{2}\left[\cos\left(\frac{\pi}{4}\right) - i\sin\left(\frac{\pi}{4}\right)\right]$$

$$a_n = d_1(1+i)^n + d_2(1-i)^n$$

$$= d_1\left[\sqrt{2}\left\{\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right\}\right]^n + d_2\left[\sqrt{2}\left\{\cos\left(\frac{\pi}{4}\right) - i\sin\left(\frac{\pi}{4}\right)\right\}\right]^n$$

$$= d_1\left[(\sqrt{2})^n\left\{\cos\left(\frac{n\pi}{4}\right) + i\sin\left(\frac{n\pi}{4}\right)\right\}\right] + d_2\left[(\sqrt{2})^n\left\{\cos\left(\frac{n\pi}{4}\right) - i\sin\left(\frac{n\pi}{4}\right)\right\}\right]$$

[Using De-Moivre's theorem]

$$= (\sqrt{2})^n\left[k_1\cos\left(\frac{n\pi}{4}\right) + k_2\sin\left(\frac{n\pi}{4}\right)\right],$$

where $k_1 = d_1 + d_2$ and $k_2 = (d_1 - d_2)i$

$$1 = a_0 = [k_1\cos(0) + k_2\sin(0)] = k_1$$

$$2 = a_1 = \sqrt{2}\left[k_1\cos\left(\frac{\pi}{4}\right) + k_2\sin\left(\frac{\pi}{4}\right)\right] = k_1 + k_2$$

$$\Rightarrow k_2 = 1$$

Therefore the general solution of the recurrance relation is given by

$$a_n = (\sqrt{2})^n\left[\cos\left(\frac{n\pi}{4}\right) + \sin\left(\frac{n\pi}{4}\right)\right], \quad n \geq 0$$

$\Downarrow$ Alternatively

$$\begin{cases} a_0 = d_1 + d_2 = 1 \\ a_1 = d_1(1+i) + d_2(1-i) = 2 \end{cases}$$

Solving these two linear equations simultaneously, we get

$$d_1 = \frac{1}{2} - \frac{i}{2} \text{ and } d_2 = \frac{1}{2} + \frac{i}{2}$$

Thus $a_n = \left(\frac{1}{2} - \frac{i}{2}\right)(1+i)^n + \left(\frac{1}{2} + \frac{i}{2}\right)(1-i)^n, \quad n \geq 0,$

which is a real number

Using De-Moivre's theorem after simplification we obtain $a_n = (\sqrt{2})^n\left[\cos\left(\frac{n\pi}{4}\right) + \sin\left(\frac{n\pi}{4}\right)\right]$ as before.

Case III: Repeated real roots

If $c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} = 0$, with $c_0 (\neq 0), c_1, c_2, \ldots, c_k (\neq 0)$ are real constants, and r is a characteristic root of multiplicity m, where $2 \leq m \leq k$, then the part of general solution that involves the root r has the form

$$A_0 r^n + A_1 n r^n + A_2 n^2 r^n + \cdots + A_{m-1} r^{m-1} n$$
$$= (A_0 + A_1 n + A_2 n^2 + \cdots + A_{m-1} n^{m-1}) r^n,$$

where $A_0, A_1, \ldots, A_{m-1}$ are arbitrary constants.

★ Exercises:-

1. Solve the following recurrance relations. [Final answer should be free of complex numbers]

(a) $a_n = 5a_{n-1} + 6a_{n-2}$, $n \geq 2$, $a_0 = 1$, $a_1 = 3$

(b) $2a_{n+2} - 11a_{n+1} + 5a_n = 0$, $n \geq 0$, $a_0 = 2$, $a_1 = -8$

(c) $a_{n+2} + a_n = 0$, $n \geq 0$, $a_0 = 0$, $a_1 = 3$

(d) $a_n - 6a_{n-1} + 9a_{n-2} = 0$, $n \geq 2$, $a_0 = 5$, $a_1 = 12$

(e) $a_n + 2a_{n-1} + 2a_{n-2} = 0$, $n \geq 2$, $a_0 = 1$, $a_1 = 3$

2. If $a_0 = 0$, $a_1 = 1$, $a_2 = 4$ and $a_3 = 37$ satisfy the recurrance relation $a_{n+2} + b a_{n+1} + c a_n = 0$, where $n \geq 0$ and b, c are constants, determine b, c and solve for a_n.

3. (a) For $n \geq 1$, let a_n count the number of binary strings of length n, where there are no consecutive 1's. Find and solve a recurrance relation for a_n.

(b) For $n \geq 1$, let b_n count the number of binary strings of length n, where there are no consecutive 1's and the first and last bit of the string are not both 1. Find and solve a recurrence relation for b_n.

4. Solve the recurrance relation $a_{n+2} = a_{n+1} a_n$, $n \geq 0$, $a_0 = 1$, $a_1 = 2$

5. (a) For $n \in \mathbb{Z}^+$, let a_n count the number of palindromes of $2n$. Then $a_{n+1} = 2a_n$, $n \geq 1$, $a_1 = 2$. Solve this first order recurrence relation for a_n.

(b) For $n \in \mathbb{Z}^+$, let b_n count the number of palindromes of $2n-1$. Set up and solve a first order recurrence relation for b_n.

6. Consider ternary strings – that is, strings where 0, 1, 2 are the only symbols used. For $n \geq 1$, let a_n count the number of ternary strings of length n where there are no consecutive 1's and no consecutive 2's. Find and solve a recurrence relation for a_n.

7. Solve the recurrance relation $a_{n+2}^2 - 5a_{n+1}^2 + 4a_n^2 = 0$, where $n \geq 0$, and $a_0 = 4$, $a_1 = 13$.

8. Determine the constants b and c if $a_n = c_1 + c_2(7^n)$, $n \geq 0$, is the general solution of the relation $a_{n+2} + ba_{n+1} + ca_n = 0$, $n \geq 0$.

9. Prove that any two consecutive fibonacci numbers are relatively prime.

The nonhomogenous recurrence relation.

Consider the case of nonhomogenous second order relation

$$a_n + c_1 a_{n-1} + c_2 a_{n-2} = kr^n,$$

where k is a constant. Here we find that the particular solution have the following cases:-

i) $a_n^{(p)} = Ar^n$, for an arbitrary constant A, if r^n is not a solution of the associated homogenous relation.

ii) $a_n^{(p)} = Bnr^n$, where B is a constant, if $a_n^{(h)} = c_1 r^n + c_2 r_1^n$, with $r_1 \neq r$;

and iii) $a_n^{(p)} = Cn^2 r^n$, for an arbitrary constant C, when $a_n^{(h)} = (c_1 + c_2 n) r^n$.

★ Given a linear nonhomogeneous recurrance relation (with constant coefficients) of the form

$$C_0 a_n + C_1 a_{n-1} + C_2 a_{n-2} + \cdots + C_k a_{n-k} = f(n),$$

where $C_0 \neq 0$ and $C_k \neq 0$, let $a_n^{(h)}$ denote the homogeneous part of the solution for a_n.

The following table may be useful.

$f(n)$	$a_n^{(p)}$
c, a constant	A, a constant
n	$A_1 n + A_0$
n^2	$A_2 n^2 + A_1 n + A_0$
$n^t, t \in \mathbb{Z}^+$	$A_t n^t + A_{t-1} n^{t-1} + \cdots + A_1 n + A_0$
$r^n, r \in \mathbb{R}$	$A r^n$
$\sin(n\theta)$	$A \sin(n\theta) + B \cos(n\theta)$
$\cos(n\theta)$	$A \sin(n\theta) + B \cos(n\theta)$
$n^t r^n$	$r^n (A_t n^t + A_{t-1} n^{t-1} + \cdots + A_1 n + A_0)$
$r^n \sin(n\theta)$	$r^n [A \cos(n\theta) + B \sin(n\theta)]$
$r^n \cos(n\theta)$	$r^n [A \cos(n\theta) + B \sin(n\theta)]$

The complete solution to the recurrance relation is given by

$$a_n = a_n^{(h)} + a_n^{(p)}$$

★ Exercises:-

1. Solve the following recurrance relations.
   (a) $a_{n+1} - a_n = 2n+3, \quad n \geq 0, \quad a_0 = 1$
   (b) $a_{n+1} - a_n = 3n^2 - n, \quad n \geq 0, \quad a_0 = 3$
   (c) $a_{n+1} - 2a_n = 5, \quad n \geq 0, \quad a_0 = 1$
   (d) $a_{n+1} - 2a_n = 2^n, \quad n \geq 0, \quad a_0 = 1$

2. Use a recurrance relation to derive the formula for $\sum_{i=0}^{n} i^2$.

3. Solve the following recurrance relations.
   (a) $a_{n+2} + 3a_{n+1} + 2a_n = 3^n, \quad n \geq 0, \quad a_0 = 0, \; a_1 = 1$
   (b) $a_{n+2} + 4a_{n+1} + 4a_n = 7, \quad n \geq 0, \quad a_0 = 1, \; a_1 = 2$
   (c) $a_{n+2} - 6a_{n+1} + 9a_n = 3(2)^n + 7(3)^n$, where $n \geq 0$. and $a_0 = 1, \; a_1 = 4$
   (d) $a_{n+3} - 3a_{n+2} + 3a_{n+1} - a_n = 3 + 5n, \quad n \geq 0$

4. Solve the following recurrance relations.
   (a) $a_{n+2}^2 - 5a_{n+1}^2 + 6a_n^2 = 7n, \quad n \geq 0, \quad a_0 = a_1 = 1$
   (b) $a_n^2 - 2a_{n-1} = 0, \quad n \geq 1, \quad a_0 = 2$ (Let $b_n = \log_2 a_n$, $n \geq 0$)

The method of generating functions

The solution of a recurrance relation can also be obtained using generating functions.

We solve following recurrance relation using generating functions.

$$a_{n+2} - 5a_{n+1} + 6a_n = 2, \quad n \geq 0, \quad a_0 = 3, \quad a_1 = 7$$

Multiply the recurrance relation by z^{n+2} and summing over all n from 0 to ∞ [* multiply by the highest subscript of recurrance relation]

$$\sum_{n=0}^{\infty} a_{n+2} z^{n+2} - 5 \sum_{n=0}^{\infty} a_{n+1} z^{n+2} + 6 \sum_{n=0}^{\infty} a_n z^{n+2} = 2 \sum_{n=0}^{\infty} z^{n+2}$$

$$\Rightarrow \left(f(z) - a_0 - a_1 z\right) - 5z\left(f(z) - a_0\right)$$

$$+ 6z^2 f(z) = 2z^2 \frac{1}{1-z}$$

$$\left[\text{assumming } f(z) = \sum_{i=0}^{\infty} a_i z^i\right]$$

$$\Rightarrow \left(f(z) - 3 - 7z\right) - 5z\left(f(z) - 3\right) + 6z^2 f(z) = \frac{2z^2}{1-z}$$

$$\Rightarrow f(z)\left[1 - 5z + 6z^2\right] = 3 - 8z + \frac{2z^2}{1-z} = \frac{3 - 11z + 10z^2}{1-z}$$

$$\Rightarrow f(z) = \frac{(3-5z)(1-2z)}{(1-5z+6z^2)(1-z)} = \frac{(3-5z)\cancel{(1-2z)}}{(1-3z)\cancel{(1-2z)}(1-z)}$$

$$\Rightarrow f(z) = \frac{3-5z}{(1-3z)(1-z)}$$

Using partial-fraction, one can get

$$f(z) = \frac{2}{1-3z} + \frac{1}{1-z} = 2\sum_{n=0}^{\infty}(3z)^n + \sum_{n=0}^{\infty} z^n$$

Taking the coefficient of z^n, we obtain

$$a_n = 2(3)^n + 1, \quad n \geq 0$$

★ System of recurrance relations can also be solved using generating functions, see the following example:

$$a_{n+1} = 2a_n + b_n,$$
$$b_{n+1} = a_n + b_n \quad \text{with } a_0 = 1,\ b_0 = 0$$

Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$ and $g(z) = \sum_{n=0}^{\infty} b_n z^n$ be the generating functions for the sequence $\{a_n\}_{n\geq 0}$ and $\{b_n\}_{n\geq 0}$, respectively.

Multiplying both the equations by z^{n+1}, we obtain

$$a_{n+1} z^{n+1} = 2a_n z^{n+1} + b_n z^{n+1}$$
$$b_{n+1} z^{n+1} = a_n z^{n+1} + b_n z^{n+1}$$

Now adding them seperately for n from 0 to ∞, one can obtain the following relations.

$$f(z) - a_0 = 2zf(z) + g(z)$$
$$g(z) - b_0 = zf(z) + zg(z)$$

$$\Rightarrow \quad f(z) - 1 = 2zf(z) + g(z)$$
$$g(z) = zf(z) + zg(z)$$

Solving these system of equations simultaneously, we obtain

$$f(z) = \frac{1-z}{z^2-3z+1} = \left(\frac{5+\sqrt{5}}{10}\right)\left(\frac{1}{\alpha - z}\right) + \left(\frac{5-\sqrt{5}}{10}\right)\frac{1}{(\beta - z)}$$

$$g(z) = \frac{z}{z^2-3z+1} = \left(\frac{-5-3\sqrt{5}}{10}\right)\left(\frac{1}{\alpha - z}\right) + \left(\frac{-5+3\sqrt{5}}{10}\right)\left(\frac{1}{\beta - z}\right),$$

where $\alpha = \frac{3+\sqrt{5}}{2}$, $\beta = \frac{3-\sqrt{5}}{2}$

This gives

$$a_n = \left(\frac{5+\sqrt{5}}{10}\right)\left(\frac{3-\sqrt{5}}{2}\right)^{n+1} + \left(\frac{5-\sqrt{5}}{10}\right)\left(\frac{3+\sqrt{5}}{2}\right)^{n+1}$$

$$b_n = \left(\frac{-5-3\sqrt{5}}{10}\right)\left(\frac{3-\sqrt{5}}{2}\right)^{n+1} + \left(\frac{-5+3\sqrt{5}}{10}\right)\left(\frac{3+\sqrt{5}}{2}\right)^{n+1},$$

$n \geq 0$.

★ Exercises :-

1. Solve the following recurrance relations by the method of generating functions.

(a) $a_{n+1} - a_n = 3^n$, $n \geq 0$, $a_0 = 1$

(b) $a_{n+1} - a_n = n^2$, $n \geq 0$, $a_0 = 1$

(c) $a_{n+2} - 3a_{n+1} + 2a_n = 0$, $n \geq 0$, $a_0 = 1$, $a_1 = 6$

(d) $a_{n+2} - 2a_{n+1} + a_n = 2^n$, $n \geq 0$, $a_0 = 1$, $a_1 = 2$

2. Solve the following systems of recurrance relations.

(a) $a_{n+1} = -2a_n - 4b_n$, $b_{n+1} = 4a_n + 6b_n$, $n \geq 0$, $a_0 = 1$, $b_0 = 0$

(b) $a_{n+1} = 2a_n - b_n + 2$, $b_{n+1} = -a_n + 2b_n - 1$, $n \geq 0$, $a_0 = 0$, $b_0 = 1$